Applied Statistical Inference Coursework

March 8, 2024

The primary goal of this study is to assess the impact of an artificial stimulant on heart rate, considering the influence of body mass index (BMI) on this effect. We aim to:

- Compare the effectiveness of a linear model versus a Gamma Generalized Linear Model (GLM) for analyzing the data.
- Estimate the treatment effect of the stimulant versus placebo on heart rate, adjusting for BMI.

1 Preliminary Data Analysis

The dataset contains the following variables:

- `rest_pulse`: Resting heart rate (beats per minute), as recorded on the first day of the study.
- `stimulated_pulse` Stimulated heart rate (beats per minute), as recorded on the second day of the study (after the treatment/placebo was administered).
- `bmi`: Body Mass Index (kg/m^2).
- `treatment`: Indicator for the treatment group (0 = stimulant, 1 = placebo).

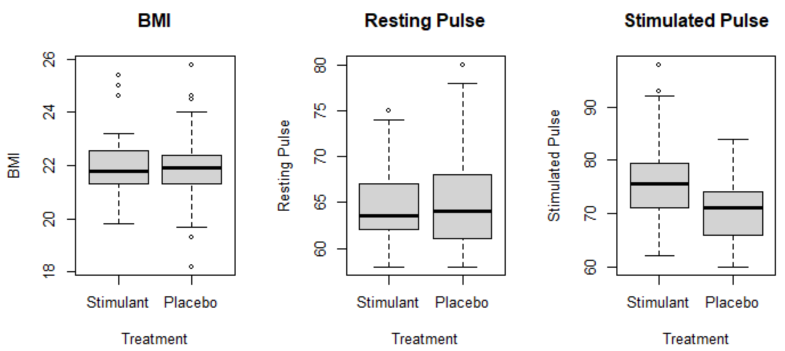


Figure 1: Data Boxplots

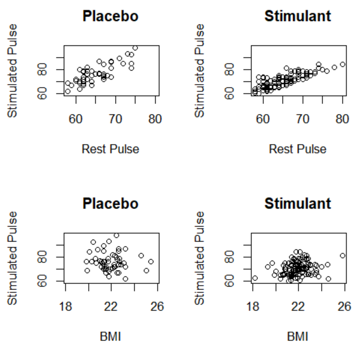


Figure 2: Data Scatter Plots

There are 118 placebo and 56 stimulant observations, which seems to be a reasonable sample size. But the data is not balanced, since the ratio of placebo to stimulant observations is 2.1. A better split may be 50% for each group, which would make the data more balanced and the results more reliable.

Looking at Figure 1, we can see that the data is evenly distributed between the two treatment groups, as the boxplots for the predictors are similar in both groups. Moreover, we we can see that the stimulated pulse tends to be higher for the stimulant group compared to the placebo group. This suggests that the treatment might have an effect on the stimulated pulse, but further analysis must be conducted to confirm this.

Looking at Figure 2, we can see that the stimulated pulse is usually higher than the rest pulse and the relationship between them seems to be linear. This is an indication that the rest pulse might be a good predictor of the stimulated pulse. On the other hand, there is no clear relationship between the stimulated pulse and BMI. This suggests that BMI might not be a good predictor of the stimulated pulse, but further analysis must be conducted to confirm this.

2 Models

In order to assess the impact of the stimulant on heart rate, we fit two models:

Linear Regression Model (*initially suggested by the clinician*): The initial model proposed involves using a linear regression framework to explain the stimulated heart rate based on the resting heart rate and treatment group. This model implicitly assumes:

- **Linearity**: The relationship between the stimulated heart rate and both the resting heart rate and treatment group is linear.
- **Independence**: Observations (heart rates) are independent of each other.
- **Homoscedasticity**: The variance of the error terms is constant across all levels of the independent variables.
- **Normality**: The error terms are normally distributed.

This model uses the least squares method, which minimizes the sum of squared differences between observed and predicted values. By using matrix algebra, we can find the coefficients that minimize the sum of squared differences between observed and predicted values.

Gamma Generalized Linear Model (GLM) with Inverse Link (*proposed by the data analyst*): To address the limitations of the linear model and better reflect the nature of the data, a Gamma GLM with an inverse link function was recommended. This model was chosen to account for the positive skewness observed in heart rate data and to model the difference between stimulated and resting heart rates as a function of BMI and treatment. The assumptions for this model include:

- **Distribution of Response Variable**: The difference between stimulated and resting heart rates follows a Gamma distribution, which is appropriate for modeling positive continuous data with skewness.
- **Mean-Variance Relationship**: The variance of the response variable is a function of its mean, which is a characteristic of the Gamma distribution. This allows for more flexibility compared to the constant variance assumption in linear regression.
- **Independence**: Observations (heart rates) are assumed to be independent of each other.
- **Link Function**: The inverse link function is used to model the mean of the response variable as a function of the linear predictors. This is appropriate for modeling positive continuous data. This choice of link function is suitable for data where the effect of predictors on the response variable is multiplicative and inversely related.

In order to fit the GLM algorithm, we're going to use the Iterated Weighted Least Squares (IWLS) algorithm:

1. Instead of initializing the regression coefficients (as typically done by the algorithm), we **initialize the mean** μ_0 to a vector consisting of the mean of the outputs y_i. We do that in order to pick a good starting point for the algorithm, which is important for the convergence of the algorithm.
2. For this model, we have the link function $\eta = \frac{1}{\mu}$, hence $\frac{\partial \eta}{\partial \mu} = -\frac{1}{\mu^2}$.
3. **The adjusted dependent variable** (z_i) for weighted regression is then:
$$z_i = \eta_i + (y_i - \mu_i) \left.\frac{\partial \eta}{\partial \mu}\right|_{\mu=\mu_i} = \eta_i - \frac{y_i - \mu_i}{\mu_i^2}.$$
4. **The weight** (w_i) for weighted regression is then:
$$w_{ii}^{-1} = \left(\frac{\partial \eta}{\partial \mu}\right)^2 V(\mu) = \frac{\mu^2}{\mu^4} = \frac{1}{\mu^2}$$
That is because the Gamma distribution is an exponential family with canonical parameter $\theta = \frac{-\beta}{\alpha}$, dispersion parameter $\phi = \frac{1}{\alpha}$, $b(\theta) = -\log(-\theta)$, $\mu = \frac{-1}{\theta}$, and $V(\mu) = \mu^2$.
5. **Fit the weighted regression** on the adjusted dependent variable z_i with weights w_i.
6. **Update the regression coefficients** (β) based on the weighted regression.
7. **Update the linear predictor** (η_i) and **the mean**, which are given by $\eta_i = X_i\beta$ and $\mu_i = \frac{1}{\eta_i}$.

2.1 Linear Model

The linear model fits quite well. By looking at the diagnostic plots (not shown), we notice that the errors have constant variance and there are no influential points in the data. But there is a slight issue with the normality of the residuals, which seem more right-skewed than a normal distribution. This suggests using a Gamma GLM with an inverse link function to better reflect the nature of the data.

We start by fitting the model `pulse_diff ~ bmi + treatment, family = Gamma(link = "inverse")` and construct a 95% confidence interval for the size of the treatment effect. By using Bootstrap, we estimate the size of the treatment effect to be around 5.457723 with a 95% confidence interval of $[4.029513, 6.923186]$.

2.2 GLM Model Selection

Let us try to find a better model by

```
Model 1: pulse_diff ~ bmi + treatment, family = Gamma(link = "inverse")
Model 2: pulse_diff ~ bmi, family = Gamma(link = "inverse")
Model 3: pulse_diff ~ treatment, family = Gamma(link = "inverse")
Model 4: pulse_diff ~ bmi * treatment, family = Gamma(link = "inverse")
```

After comparing the models by looking at the deviances, the AIC, and the diagnostic plots (not shown), we've found that the best model is Model 4: this model has the lowest AIC and the lowest deviance, indicating that it may be the best model to explain the data.

Notice that both the model proposed by the clinician and the model proposed by the statistician share the same flaw: They are unable to model the interaction between BMI and treatment. Intuitively, the pulse difference should be influenced by the BMI only for the stimulant group, and not for the placebo group. This is another reason why Model 4 is the best model: it includes this interaction.

2.3 Best Model

Let us analyze the best model. By looking at the diagnostic plots (not shown), we notice that the errors have constant variance, but there are a couple of influential points in the data. If we would want to, we could remove these points, but since the data is not large, we should be careful with removing observations. But in this case, we see that the errors are more normally distributed than in the linear model, which suggests that this model is an improvement over the linear model.

Coefficient	**Estimate**	**Std. Error**	t **value**	**Pr**$(>\|t\|)$
(Intercept)	0.049042	0.109212	0.449	0.654
bmi	0.001771	0.004992	0.355	0.723
treatment1	0.788984	0.165805	4.759	4.15e-06 ***
bmi:treatment1	-0.031821	0.007466	-4.262	3.35e-05 ***

Table 1: GLM Coefficients

Looking at Table 1, we can see that `treatment1` and `bmi:treatment1` are the only statistically significant coefficients. This suggests that the treatment has a significant effect on the stimulated pulse, and that the BMI has a significant effect on the stimulated pulse only for the stimulant group, which aligns with our intuition.

The significance of the coefficient `bmi:treatment1` indicates that the BMI might indeed have an effect on the stimulated pulse in the treatment group, making it a good predictor for the stimulated pulse in the case of administered treatment.

Coefficient	Estimate	95% Confidence Interval
(Intercept)	0.049	[-0.169, 0.257]
bmi	0.002	[-0.008, 0.012]
treatment1	0.789	[0.457, 1.106]
bmi:treatment1	-0.032	[-0.046, -0.017]

Table 2: 95% Confidence Intervals for the coefficients

The confidence intervals for the coefficients are shown in Table 2. They were built using the `confit()` function in R, which in this case of a Gamma GLM with an inverse link function, uses the profile likelihood method to construct the confidence intervals. This approach evaluates the likelihood of the parameter estimates across a range of values, identifying the interval within which the likelihood of the estimates does not decrease significantly from its maximum value. This method does not rely on the normal approximation, making it more accurate for distributions like Gamma, especially with non-linear link functions such as the inverse link.

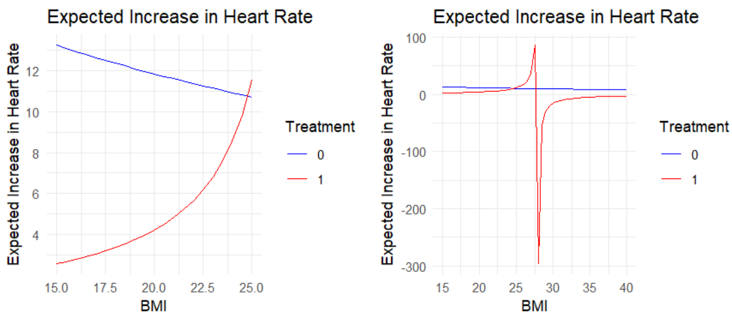


Figure 3: Expected Increase in Heart Rate

Figure 3 shows the expected increase in heart rate by varying the BMI for both groups.
In the first plot, we supply the model with a range of BMI values similar to the one in the observed data, and we can see that the expected increase in heart rate is higher for the stimulant group compared to the placebo group for the most common values (that is, if we're not including the outliers, which can be seen in Figure 1). This suggests that, in the given BMI range, the stimulant has a significant effect on the stimulated pulse.
In the second plot, we supply the model with a larger range of BMI values which are not present in the observed data, and we can see something interesting happening: the predictions for the pulse difference explode as the BMI increases. The trend is caused by the inverse link function being supplied with values close to 0, resulting in an explosion of the predictions. This suggests that the model is not reliable for larger BMI values outside the range of the observed data, so more data should be collected for these values.

Figure 4 shows the power of different sizes of treatment effect. We see that, as the treatment effect increases, the power of the test increases as well. This makes sense: the larger the treatment effect, the easier it is to detect it. After around 1.6, the power is over 0.05, which means that from that point onwards, there are chances of detecting the treatment effect. The standard threshold for adequate power is 0.8, which seems to not be achieved by any treatment effect from the range $[0.1, 20]$. But if we consider 0.6 as an adequate power, then the treatment effect can be reliabily detected if its value is larger than 10.9.

3 Conclusion

Here are some important conclusions from this analysis:

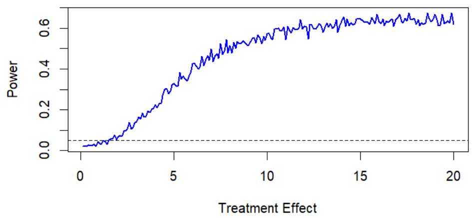


Figure 4: Power for different treatment effects

- The data is not balanced, since the ratio of placebo to stimulant observations is 2.1. In order to make the results more reliable, the data should be balanced, with 50% of the observations in each group.
- The GLM model does not generalize well for BMI values which are larger than the ones in the observed data, so more data should be collected for individuals with large BMI values.
- In the case the stimulant is administered, the BMI influences the stimulated pulse.

4 Asymptotic approximations

We need to check the following asymptotic approximations:

1. The **MLE is asymptotically normal**, which means that for large sample sizes, the MLE is approximately normally distributed. This is important because it allows us to construct confidence intervals and perform hypothesis tests using the normal distribution as an approximation for the MLE.
2. The **scaled deviances are asymptotically χ^2-distributed** with degrees of freedom equal to the difference in the number of parameters between the full and reduced models.

Coefficient	**p-value**
Intercept	0.7410
bmi	0.6945
treatment1	0.3199
bmi:treatment1	0.2775

Figure 5: Kolmogorov-Smirnov test p-values for coefficients

We're going to check these assumptions by running a simulation study, with 1000 iterations.

By looking at Figure 6, we can see that the coefficients are approximately normally distributed, which suggests that the MLE is asymptotically normal. Moreover, by looking at Table 5, we can see that the p-values for the Kolmogorov-Smirnov test are all larger than 0.05, which suggests that the coefficients are normally distributed.
Let's also try a multivariate normal test from the `MVN` package in R. We're going to use the Mardia test. The p-value of the Mardia skewness test is approximately $6.54 - 30$, while the p-value of the Mardia kurtosis test is approximately 0.00011. Both of these values are smaller than 0.05, which suggests that the data is not multivariate normal. Further analysis should be conducted to confirm this, and, in the case of non-normality, to find the cause of it.

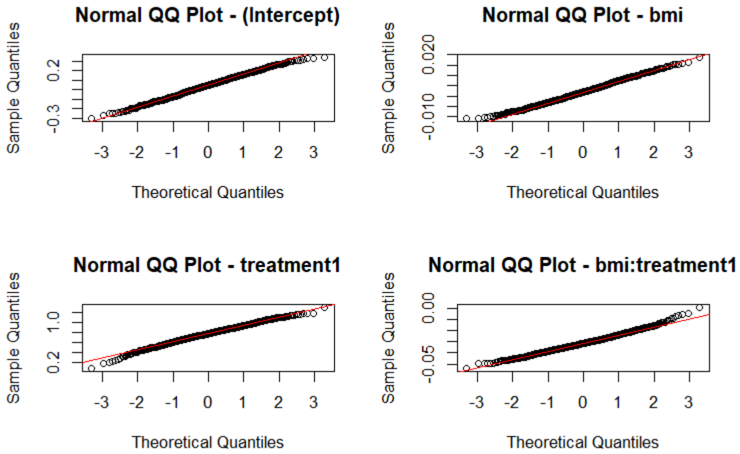


Figure 6: QQ plots for the coefficients

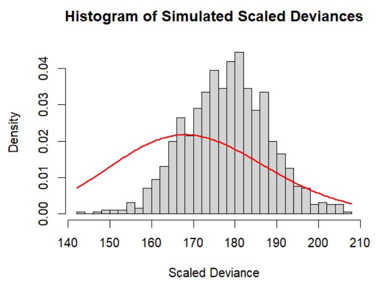


Figure 7: Histogram of the scaled deviances against the χ^2 distribution

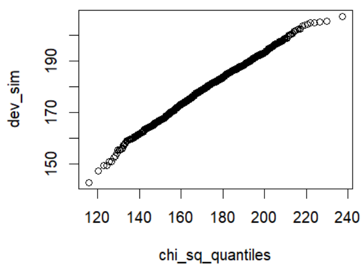


Figure 8: QQ plot of the scaled deviances against the χ^2 distribution

Now let's check the scaled deviances. By looking at Figure 7 and Figure 8, we can see that the scaled deviances don't seem to be χ^2-distributed. They seem to be off by a translation and a scaling factor. Further analysis should be conducted to confirm this, and, in the case of non-χ^2-distribution, to find the cause of it.

Although the non-χ^2-distribution of the scaled deviances can be noticed from the plots, we conduct a Kolmogorov-Smirnov test to confirm this. The p-value of the test is smaller than $2.2e-16$, which suggests that the scaled deviances are not χ^2-distributed.

5 Comparison with another study

Let us start by analyzing our data.

Figure 9 shows no sign of a clear relationship between BMI and resting pulse rate. Next, we compute the Pearson correlation coefficient between the two variables, which is -0.028. This indicates a very weak negative linear relationship between the two variables, but is likely not practically significant.

Overall, there seems to be no clear relationship between the two variable, whici suggests that BMI might not be a good predictor for the resting pulse.

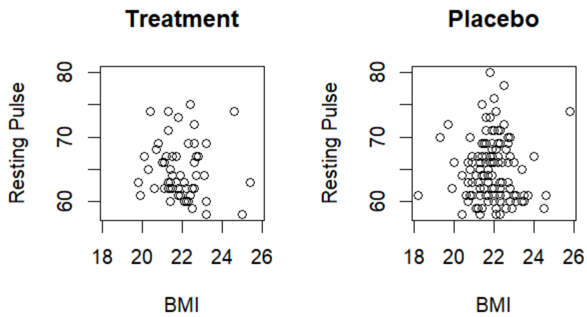


Figure 9: Resting Pulse vs BMI

Not let's us compare our results with the results of another study:
https://bmccardiovascdisord.biomedcentral.com/articles/10.1186/s12872-019-01286-2

This study investigates the interaction between body mass index (BMI), exercise capacity, and the variance in resting heart rate (ΔRHR) over time. It emphasizes BMI and exercise capacity as significant predictors of ΔRHR, proposing that these factors could help identify individuals at increased cardiovascular risk efficiently and cost-effectively.

Findings indicate that a higher BMI and reduced exercise capacity are linked to an uptick in resting heart rate, suggesting the need for lifestyle modifications or preventive strategies to mitigate cardiovascular risk.

This contradicts the previous analysis of our data, which suggested that BMI might not be a good predictor for the resting pulse. One of the causes for this discrepancy might be the difference in the sample size: the study has a sample size of over $n = 6000$, while our sample size is $n = 174$. This suggests that the results of the study might be more reliable than the results of our analysis.

A Appendix: Data Review

After reviewing the data, we have found the following issues:

- In order to get more reliable results, the data should be balanced, with 50% of the observations in each group. Our current ratio of placebo to stimulant observations is 2.1, which clearly does not respect this condition.
- Our model fits well for BMI values which are similar to the ones in the observed data, but it does not generalize well for larger BMI values. More data should be collected from individuals with large BMI values, since the stimulant is likely to have a different effect on them.
- In the provided data, there is a notable increase in heart rate with the stimulant, one our estimation being around 5.457723 with a 95% confidence interval of $[4.029513, 6.923186]$.
- In our simulation study, we've found out that in order to be at least 60% confident in detecting the treatment effect, the treatment effect should be larger than 10.9. This suggests that the treatment effect is not easily detectable with the current sample size, and more data should be collected in order to improve the reliability of our results.
- The initial model is unable to capture the interaction between the BMI and the treatment: the pulse difference should be influenced by the BMI only for the stimulant group, and not for the placebo group.

Appendix: R Code

```r
# Import the necessary libraries
library(boot)
library(ggplot2)
library(gridExtra)
library(MASS)
if (!require("MVN")) {
  install.packages("MVN")
}
```

```
## Loading required package: MVN
```

```
## Warning: package 'MVN' was built under R version 4.3.3
```

```r
library(MVN)
```

1 Load the data and extract the variables

```r
# Load the data
load("1854554.RData")

# Tell R that treatment is categorical
dat$treatment <- as.factor(dat$treatment)
# Add the pulse difference to the data frame
dat$pulse_diff = dat$stimulated_pulse - dat$rest_pulse

# Attach the data frame
attach(dat)
```

```r
# View the data
# View(dat)
```

2 Exploratory analysis of the data

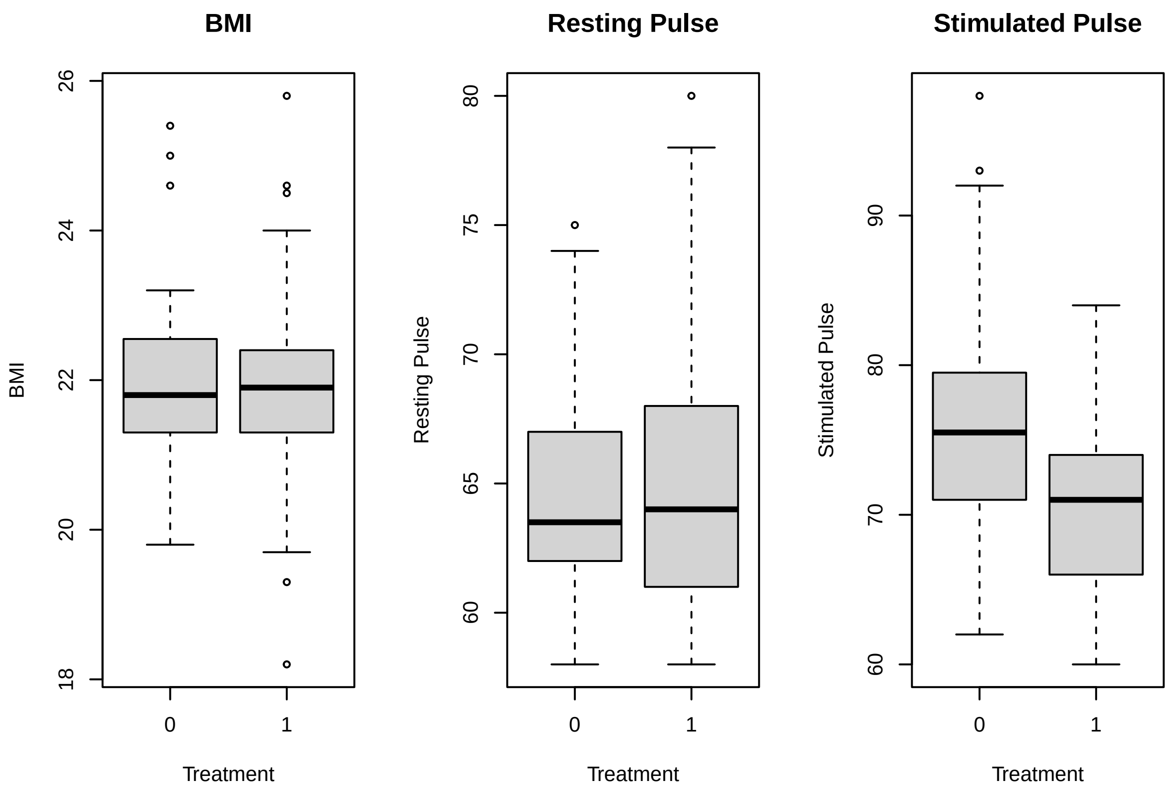
```r
# Print the first few rows of the data
head(dat)
```

```
##   rest_pulse stimulated_pulse  bmi treatment pulse_diff
## 1         58               62 23.2         0          4
## 2         67               71 22.2         1          4
## 3         68               75 20.7         0          7
## 4         66               68 22.0         1          2
## 5         61               64 23.7         1          3
## 6         66               73 21.8         1          7
```

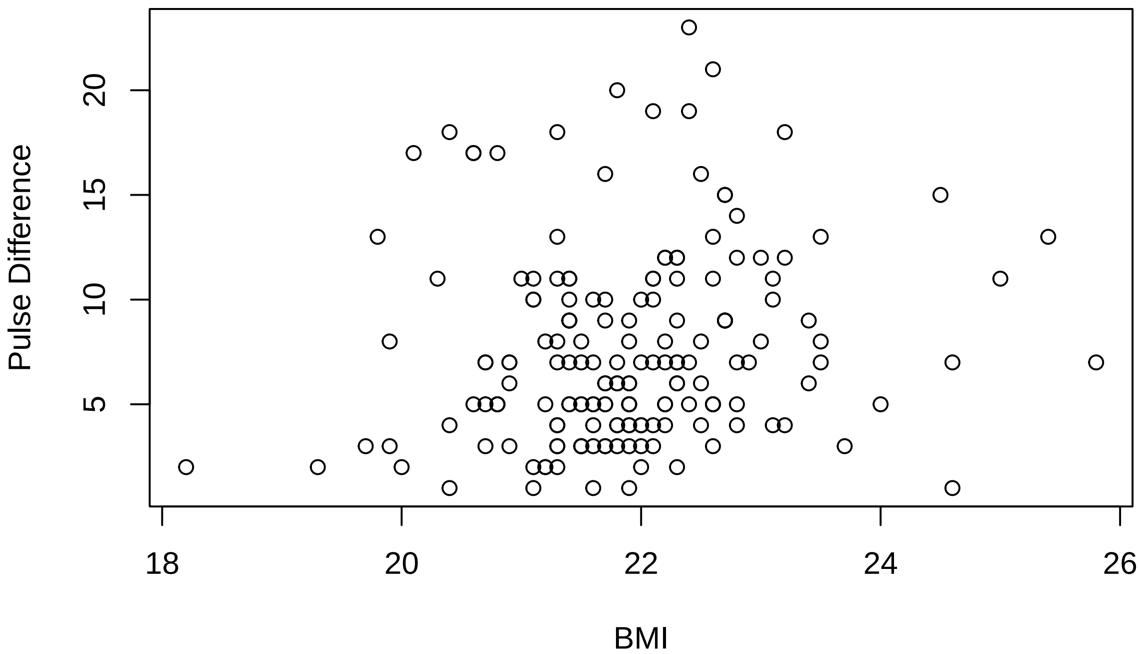
```r
# Summary of the data
summary(dat)
```

```
##     rest_pulse     stimulated_pulse       bmi         treatment    pulse_diff
##  Min.   :58.00   Min.   :60.00    Min.   :18.20   0: 56      Min.   : 1.00
##  1st Qu.:61.00   1st Qu.:67.00    1st Qu.:21.30   1:118      1st Qu.: 4.00
##  Median :64.00   Median :72.00    Median :21.90              Median : 7.00
##  Mean   :64.91   Mean   :72.49    Mean   :21.90              Mean   : 7.58
##  3rd Qu.:67.00   3rd Qu.:76.00    3rd Qu.:22.48              3rd Qu.:10.00
##  Max.   :80.00   Max.   :98.00    Max.   :25.80              Max.   :23.00
```

```r
par(mfrow = c(1, 3))

# Box plot for BMI by Treatment
boxplot(bmi ~ treatment, main = "BMI",
        xlab = "Treatment", ylab = "BMI")

# Box plot for Resting Pulse by Treatment
boxplot(rest_pulse ~ treatment, main = "Resting Pulse",
        xlab = "Treatment", ylab = "Resting Pulse")

# Box plot for Stimulated Pulse by Treatment
boxplot(stimulated_pulse ~ treatment, main = "Stimulated Pulse",
        xlab = "Treatment", ylab = "Stimulated Pulse")
```



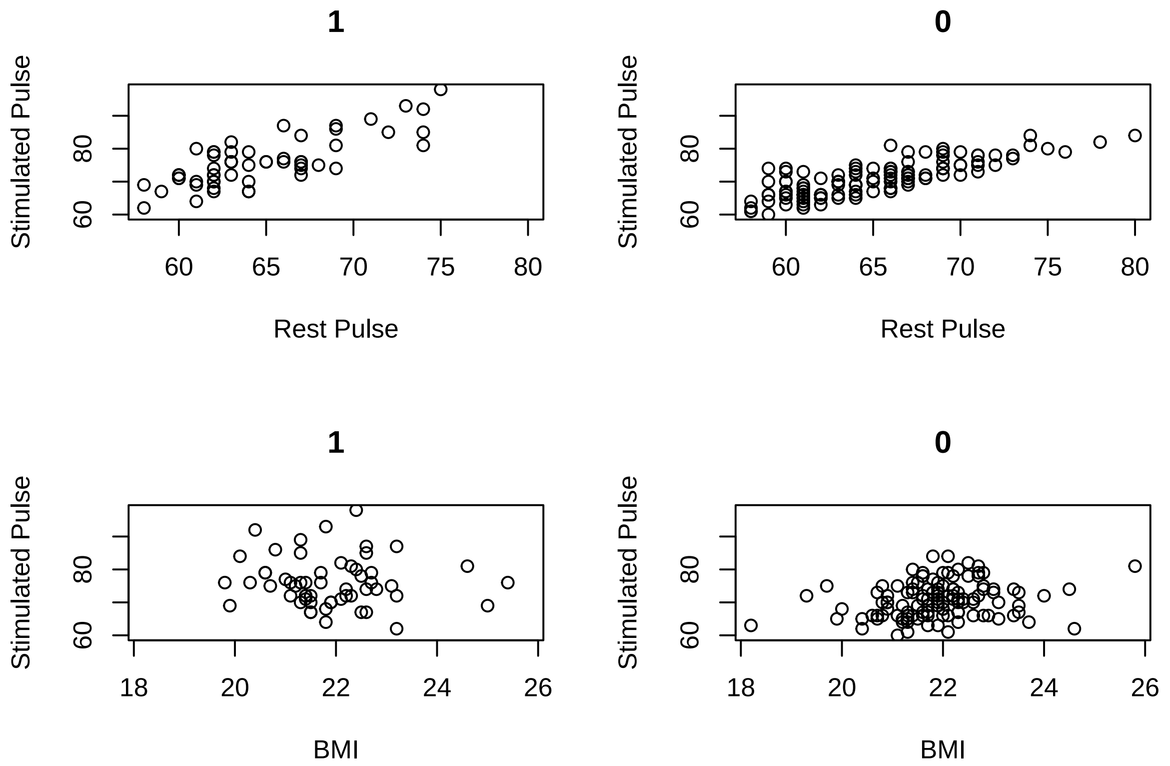

```r
# Plot the pulse difference
plot(pulse_diff ~ bmi, xlab = "BMI", ylab = "Pulse Difference")
```




```
remove_outliers <- function(df, column_name) {
  Q1 <- quantile(df[[column_name]], 0.25)
  Q3 <- quantile(df[[column_name]], 0.75)
  IQR <- Q3 - Q1
  lower_bound <- Q1 - 1.5 * IQR
  upper_bound <- Q3 + 1.5 * IQR
  df <- df[df[[column_name]] >= lower_bound & df[[column_name]] <= upper_bound, ]
  return(df)
}

# Remove the outliers
# dat <- remove_outliers(dat, "bmi")
# dat <- remove_outliers(dat, "rest_pulse")
# dat <- remove_outliers(dat, "stimulated_pulse")
# Attach the data frame
# attach(dat)
```

```
# Set up the plot area
par(mfrow = c(2, 2))

# Scatter plots for the rest pulse
plot(rest_pulse[treatment == 0], stimulated_pulse[treatment == 0],
     xlab = "Rest Pulse", ylab = "Stimulated Pulse",
     main = 1, xlim = range(rest_pulse), ylim = range(stimulated_pulse))
plot(rest_pulse[treatment == 1], stimulated_pulse[treatment == 1],
     xlab = "Rest Pulse", ylab = "Stimulated Pulse",
```

```r
     main = 0, xlim = range(rest_pulse), ylim = range(stimulated_pulse))

# Scatter plots for the BMI
plot(bmi[treatment == 0], stimulated_pulse[treatment == 0],
     xlab = "BMI", ylab = "Stimulated Pulse",
     main = 1, xlim = range(bmi), ylim = range(stimulated_pulse))
plot(bmi[treatment == 1], stimulated_pulse[treatment == 1],
     xlab = "BMI", ylab = "Stimulated Pulse",
     main = 0, xlim = range(bmi), ylim = range(stimulated_pulse))
```



3 Initial model suggested by the clinician

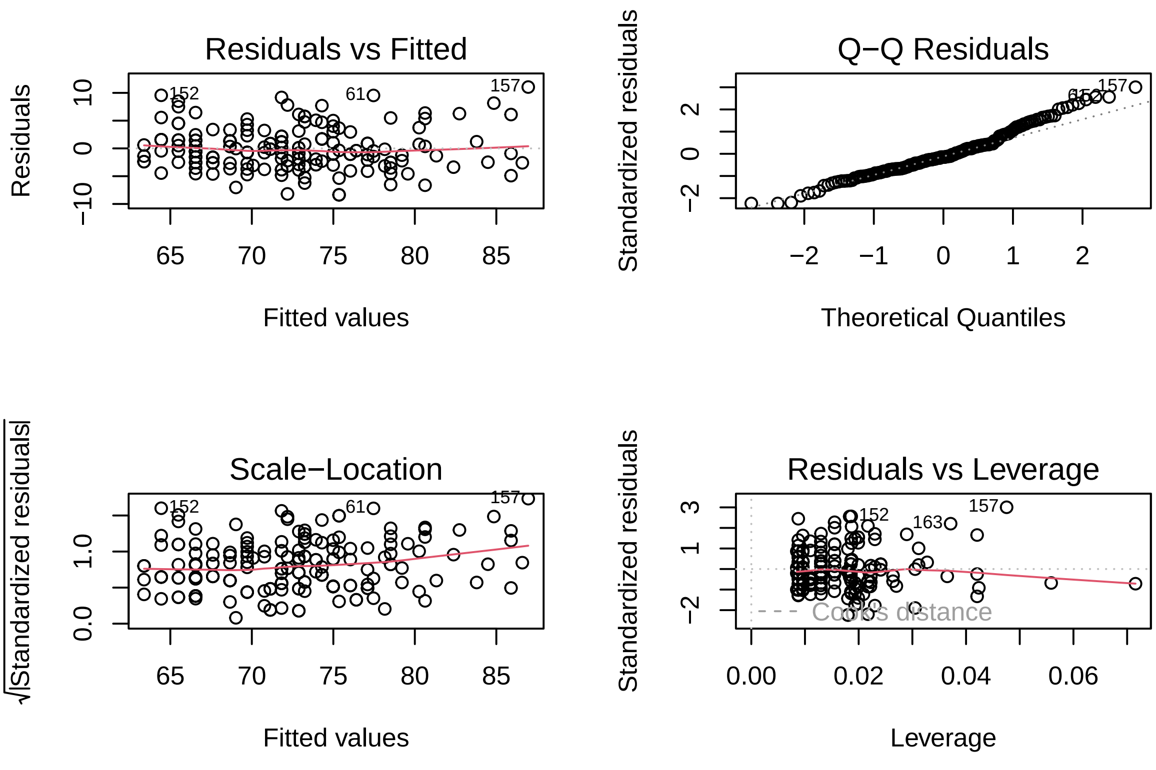
```r
# Fit the initial linear model
fit0 <- lm(stimulated_pulse ~ rest_pulse + treatment)
```

```r
# Set up the plot area
par(mfrow = c(2, 2))

# Summary of the initial model
summary(fit0)
```

```
##
## Call:
## lm(formula = stimulated_pulse ~ rest_pulse + treatment)
##
```

```
## Residuals:
##      Min       1Q  Median       3Q      Max
## -8.3537 -2.5430 -0.5512  1.6656 11.0426
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.8412     4.1143   1.906   0.0584 .
## rest_pulse    1.0549     0.0631  16.718   <2e-16 ***
## treatment1   -5.6378     0.6114  -9.221   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.766 on 171 degrees of freedom
## Multiple R-squared:  0.6752, Adjusted R-squared:  0.6714
## F-statistic: 177.7 on 2 and 171 DF,  p-value: < 2.2e-16
```

```
# Plot the initial model
par(mfrow = c(2, 2))
plot(fit0)
```



4 Model suggested by the statistician

```
custom_glm_model <-
  function(X,
           y,
```

```r
          max_iter = 100,
          tol = 1e-4,
          eps = 1e-6) {
    # Initialize mu and eta
    mu <- rep(mean(y), length(y))
    eta <- 1 / mu

    # Initialize beta
    beta <- rep(0, ncol(X))

    for (i in 1:max_iter) {
      # Update the adjusted dependent variable z
      z <- eta - ((y - mu) / mu ^ 2)

      # Update the weights
      w <-  mu ^ 2

      # Update the model using a weighted least squares fit
      WLS_fit <- lm(z ~ bmi + treatment, weights = w)
      new_beta <- as.numeric(WLS_fit$coeff)

      # Check if the coefficients have converged
      if (max(abs(new_beta - beta)) < tol) {
        return(new_beta)
      }

      # Update the coefficients
      beta <- new_beta

      # Update the linear predictor
      eta <- X %*% beta

      # Update the mean
      mu <- 1 / eta
    }

    return (beta)
  }

X <- cbind(1, bmi, treatment)
y <- pulse_diff

# Fit the custom GLM model
coeffs <- custom_glm_model(X, y, max_iter=1000)
coeffs
```

```
## [1] -0.049360490  0.004422611  0.094761065
```

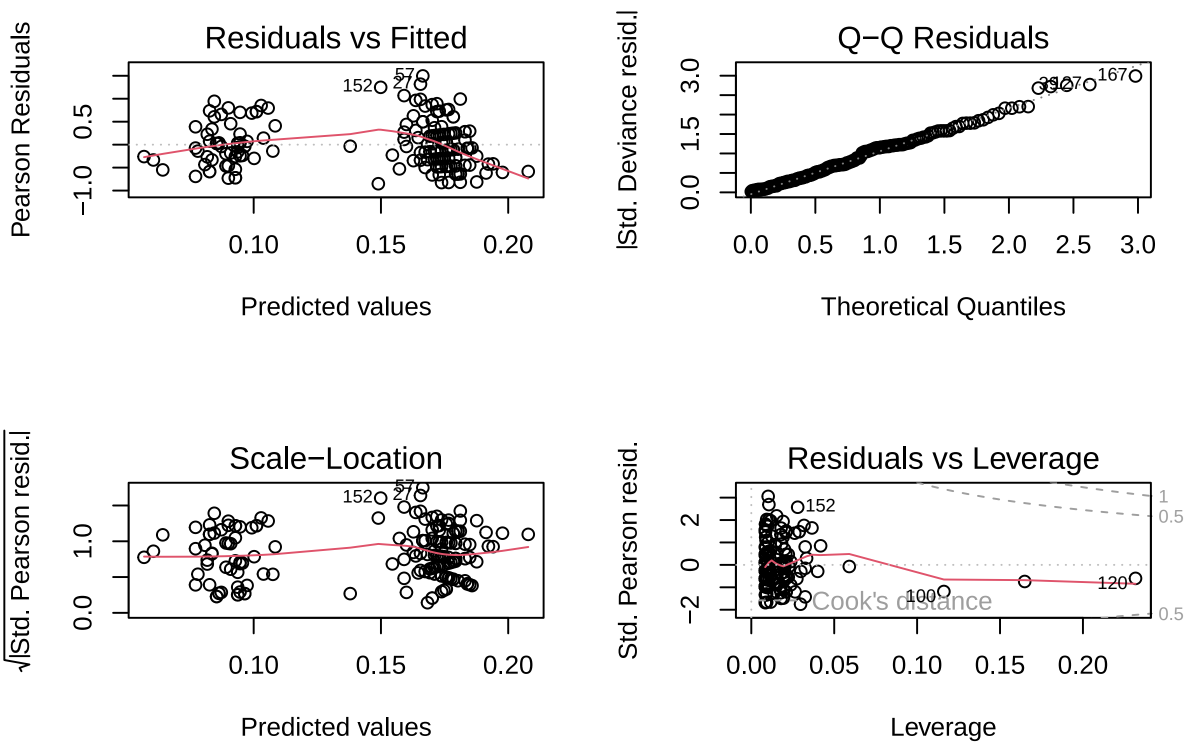
The algorithm converges pretty rapidly and we get similar coefficients to the ones obtained by the R's `glm` function. Hence, from now on, we will use the `glm` function to fit the models.

```r
# Fit the GLM model
fit1_form <- (pulse_diff ~ bmi + treatment)
fit1 <- glm(fit1_form, family = Gamma(link = "inverse"), data = dat)
```

```r
# Summary of the GLM model
summary(fit1)
```

```
##
## Call:
## glm(formula = fit1_form, family = Gamma(link = "inverse"), data = dat)
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.290659   0.085576   3.397 0.000849 ***
## bmi         -0.009202   0.003843  -2.395 0.017707 *
## treatment1   0.084694   0.009740   8.695 2.76e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.2425243)
##
##     Null deviance: 67.433  on 173  degrees of freedom
## Residual deviance: 47.430  on 171  degrees of freedom
## AIC: 920.98
##
## Number of Fisher Scoring iterations: 5
```

```r
# Plot the GLM model
par(mfrow = c(2, 2))
plot(fit1)
```



5 Estimation of the treatment effect

```r
ci_bootstrap <- function(data, fit_form, num_iterations = 1000, seed=60012) {
  # Set the seed for reproducibility
  set.seed(seed)

  # Compute the typical BMI (mean BMI for the sample)
  typical_bmi <- mean(bmi)

  # Extract the number of points in the data
  num_points <- nrow(dat)

  # Create an empty vector to store the predicted differences
  predicted_diff <- numeric(num_iterations)

  for (i in 1:num_iterations) {
    # Sample the data with replacement
    indexes <- sample(num_points, num_points, replace = TRUE)

    # Train the model
    fit_i <- glm(fit_form, family = Gamma(link = "inverse"), data = data[indexes,])

    # Predict the difference in pulse
    new_data <- data.frame(bmi = typical_bmi,
                           treatment = factor(c(0, 1),levels = levels(treatment)))

    predictions <- predict(fit_i, newdata = new_data, type = "response")
    predicted_diff[i] <- predictions[1] - predictions[2]
  }

  # Compute the estimated difference
  estimated_diff <- mean(predicted_diff)

  # Compute the 95% confidence interval
  conf_interval <- quantile(predicted_diff, c(0.025, 0.975))

  return(list(estimated_diff = estimated_diff,
              conf_interval = conf_interval))
}

ci_bootstrap(dat, fit1_form, num_iterations = 1000)
```

```
## $estimated_diff
## [1] 5.457723
##
## $conf_interval
##     2.5%    97.5%
## 4.029513 6.923186
```

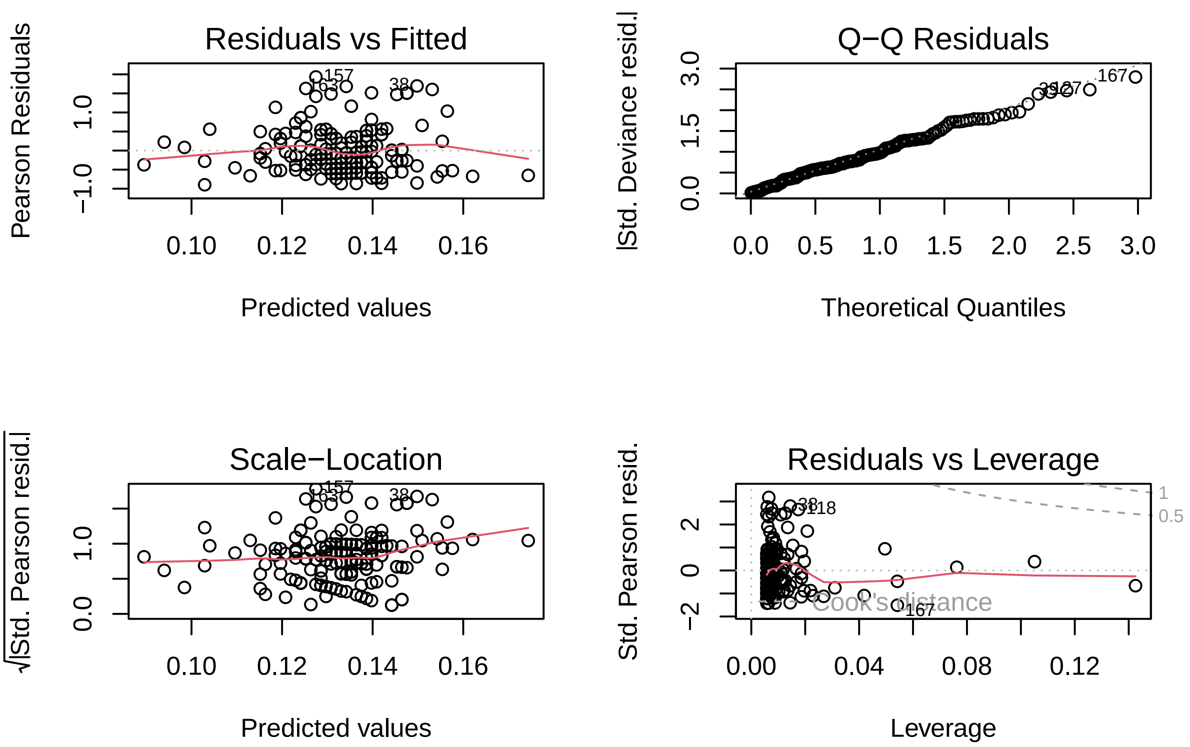
6 Changing the linear predictors

```r
# Fit the GLM model
fit2 <- glm(pulse_diff ~ bmi, family = Gamma(link = "inverse"), data = dat)
```

```r
# Summary of the GLM model
summary(fit2)
```

```
##
## Call:
## glm(formula = pulse_diff ~ bmi, family = Gamma(link = "inverse"),
##     data = dat)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.377514   0.118314   3.191  0.00169 **
## bmi         -0.011163   0.005344  -2.089  0.03818 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.3723593)
##
##     Null deviance: 67.433  on 173  degrees of freedom
## Residual deviance: 65.909  on 172  degrees of freedom
## AIC: 979.28
##
## Number of Fisher Scoring iterations: 5
```

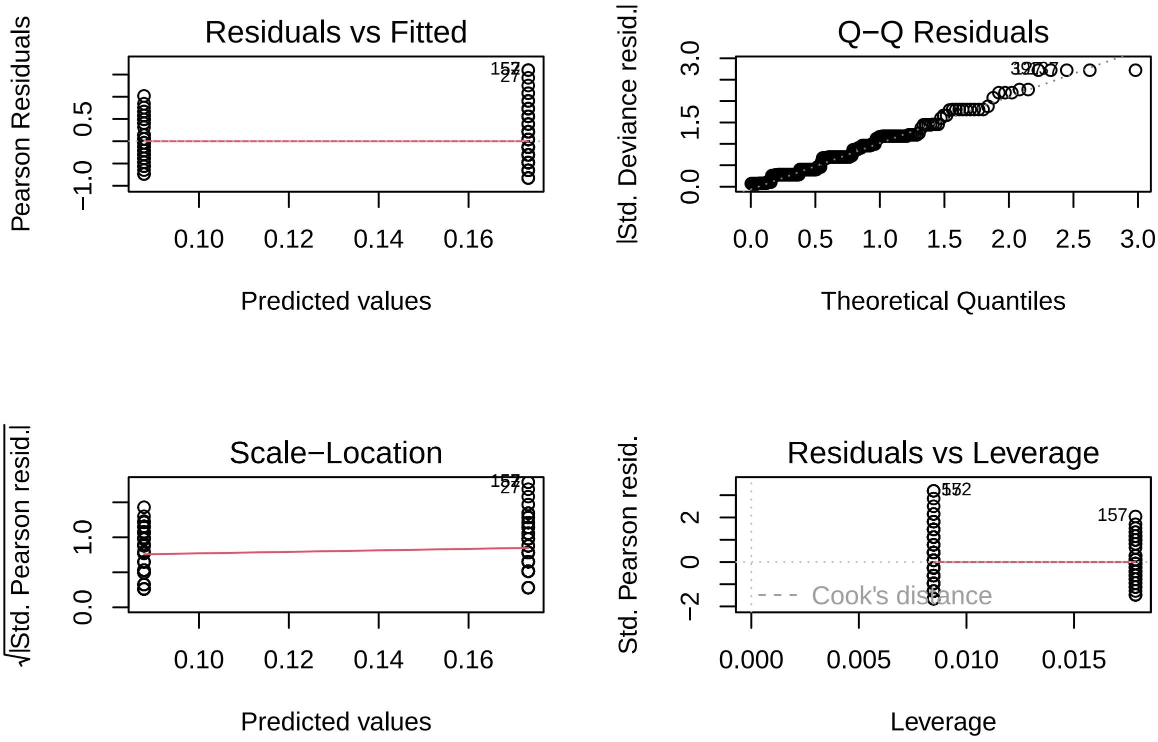
```r
# Plot the GLM model
par(mfrow = c(2, 2))
plot(fit2)
```



```r
# Fit the GLM model
fit3 <- glm(pulse_diff ~ treatment, family = Gamma(link = "inverse"), data = dat)

# Summary of the GLM model
summary(fit3)
```

```
##
## Call:
## glm(formula = pulse_diff ~ treatment, family = Gamma(link = "inverse"),
##     data = dat)
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.087774   0.005891  14.900  < 2e-16 ***
## treatment1  0.085500   0.009944   8.598 4.83e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.2522337)
##
##     Null deviance: 67.433  on 173  degrees of freedom
## Residual deviance: 48.707  on 172  degrees of freedom
## AIC: 923.81
##
## Number of Fisher Scoring iterations: 5
```

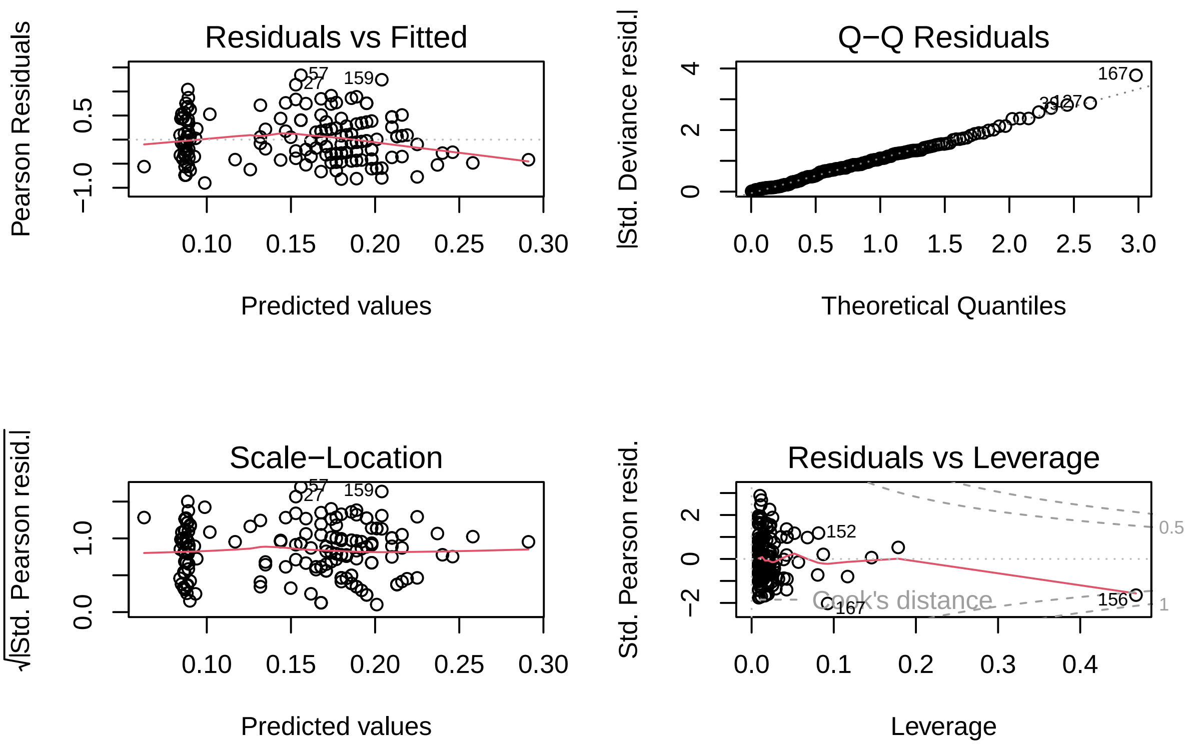
```r
# Plot the GLM model
par(mfrow = c(2, 2))
plot(fit3)
```




```r
# Fit the GLM model
fit4_form <- (pulse_diff ~ bmi * treatment)
fit4 <- glm(fit4_form, family = Gamma(link = "inverse"), data = dat)

summary(fit4)
```

```
##
## Call:
## glm(formula = fit4_form, family = Gamma(link = "inverse"), data = dat)
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     0.049042   0.109212   0.449    0.654
## bmi             0.001771   0.004992   0.355    0.723
## treatment1      0.788984   0.165805   4.759 4.15e-06 ***
## bmi:treatment1 -0.031821   0.007466  -4.262 3.35e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.2181898)
##
##     Null deviance: 67.433  on 173  degrees of freedom
## Residual deviance: 43.796  on 170  degrees of freedom
## AIC: 908.51
##
## Number of Fisher Scoring iterations: 6
```

```r
# Plot the GLM model
par(mfrow = c(2, 2))
plot(fit4)
```



Now let's compare the models.

```r
# Compare the models
anova(fit1, fit2, fit3, fit4)
```

```
## Analysis of Deviance Table
##
## Model 1: pulse_diff ~ bmi + treatment
## Model 2: pulse_diff ~ bmi
## Model 3: pulse_diff ~ treatment
## Model 4: pulse_diff ~ bmi * treatment
##   Resid. Df Resid. Dev Df Deviance
## 1       171     47.430
## 2       172     65.909 -1 -18.4786
## 3       172     48.707  0  17.2021
## 4       170     43.796  2   4.9109
```

```r
AIC(fit1, fit2, fit3, fit4)
```

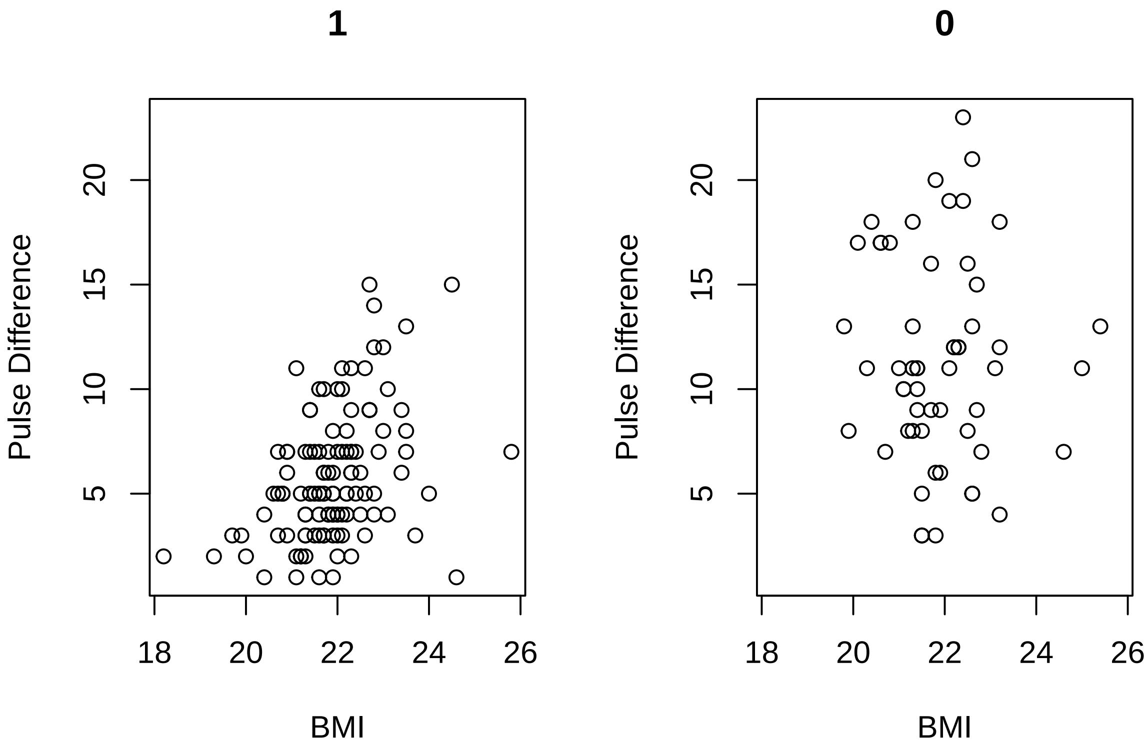
```
##      df      AIC
## fit1  4 920.9784
## fit2  3 979.2752
## fit3  3 923.8108
## fit4  5 908.5056
```

```r
# Pick the best model
fit_form <- fit4_form
fit <- fit4

# Make confidence intervals for the coefficients
confint(fit, level = 0.95)
```

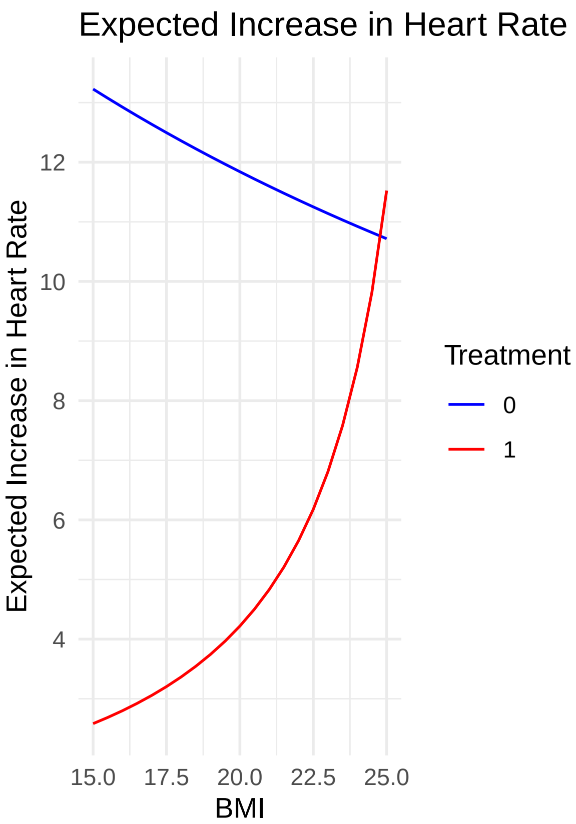
```
## Waiting for profiling to be done...

##                      2.5 %      97.5 %
## (Intercept)    -0.168640884  0.25701541
## bmi            -0.007673459  0.01178051
## treatment1      0.456602645  1.10559051
## bmi:treatment1 -0.046078094 -0.01684352
```

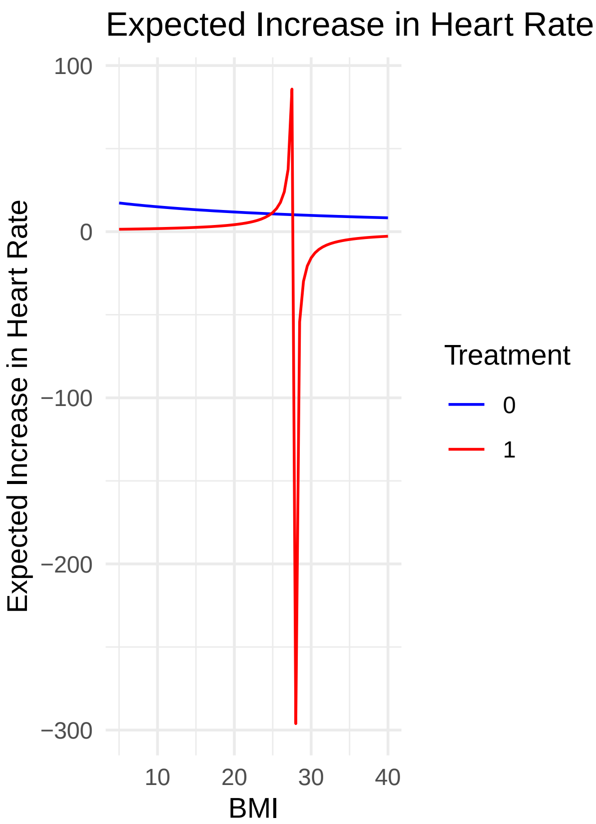
7 Expected increase in heart rate

```r
# Plot the difference in heart rate as a function of BMI
par(mfrow = c(1, 2))
plot(bmi[treatment == 1], pulse_diff[treatment == 1],
     xlab = "BMI", ylab = "Pulse Difference",
     main = 1, xlim = range(bmi), ylim = range(pulse_diff))
plot(bmi[treatment == 0], pulse_diff[treatment == 0],
     xlab = "BMI", ylab = "Pulse Difference",
     main = 0, xlim = range(bmi), ylim = range(pulse_diff))
```



```r
make_predictions <- function(fit, range) {
  # Prepare a data frame for prediction: repeat each BMI for both treatment levels
  data_stimulant = data.frame(bmi = range, treatment = factor(0, levels = levels(treatment)))
  data_placebo = data.frame(bmi = range, treatment = factor(1, levels = levels(treatment)))

  # Predict heart rate increase for the new data
  predictions_stimulant = predict(fit, newdata = data_stimulant, type = "response")
  predictions_placebo = predict(fit, newdata = data_placebo, type = "response")

  # Combine the prediction data frames into one,
  # adding a column to distinguish between stimulant and placebo
  data_combined <- rbind(data_stimulant, data_placebo)
  data_combined$treatment <- factor(data_combined$treatment, labels = c(0, 1))

  # Add the predictions as a new column in the combined data frame
  data_combined$expected_increase <- c(predictions_stimulant, predictions_placebo)

  return (data_combined)
}

plot_predictions <- function(data_combined) {
  # Use ggplot2 to plot the expected increase in heart rate as a function of BMI
  return (ggplot(data_combined, aes(x = bmi, y = expected_increase, color = treatment)) +
    geom_line() +
    labs(title = "Expected Increase in Heart Rate",
         x = "BMI",
         y = "Expected Increase in Heart Rate") +
    scale_color_manual(values = c("blue","red")) +
    theme_minimal() +
    guides(color = guide_legend(title = "Treatment")))
}

# Make predictions
predictions_1 <- make_predictions(fit, range = seq(from = 15, to = 25, by=0.5))
predictions_2 <- make_predictions(fit, range = seq(from = 5, to = 40, by=0.5))

# Plot these values on the same graph
grid.arrange(plot_predictions(predictions_1), plot_predictions(predictions_2), ncol = 2)
```





8 Simulation study to to determine the size of treatment effect

```r
# Set the seed
set.seed(5039)

num_simulations <- 600
treatment_effects <- seq(from = 0.1, to = 20, by = 0.1)

# Create an empty vector to store the results
power <- numeric(length(treatment_effects))

# Run the simulations
for (i in seq_along(treatment_effects)) {
  effect_size <- treatment_effects[i]
  insignificant_count <- 0

  # Run the simulations for the current effect size
  for (j in 1:num_simulations) {
    # Create a copy of the dataset for manipulation
    sim_data <- dat

    # Simulate the response variable
    sim_data$pulse_diff_sim <- unlist(simulate(fit))

    # Update the response variable for the placebo group
```

```
    # (we're bringing them to the same level as the stimulant group)
    # Notice that we're adding the treatment effect to the placebo group because
    # if we would subtract it from the stimulant group,
    # we would encounter errors with the Gamma model fitting.
    indexes <- sim_data$treatment == 1
    sim_data$pulse_diff_sim[indexes] <- sim_data$pulse_diff_sim[indexes] + effect_size

    # Fit the model to the simulated data
    form = pulse_diff_sim ~ bmi * treatment
    fit_sim <- glm(form, family = Gamma(link = "inverse"), data = sim_data)

    treatment_is_insignificant <- summary(fit_sim)$coefficients["treatment1", "Pr(>|t|)"] > 0.05
    bmi_treatment_interaction_is_insignificant <- summary(fit_sim)$coefficients["bmi:treatment1", "Pr(>

    # Check if the treatment effect is insignificant
    if (treatment_is_insignificant && bmi_treatment_interaction_is_insignificant) {
      insignificant_count <- insignificant_count + 1
    }
  }

  # Store the power for the current effect size
  power[i] <- insignificant_count / num_simulations
}
```

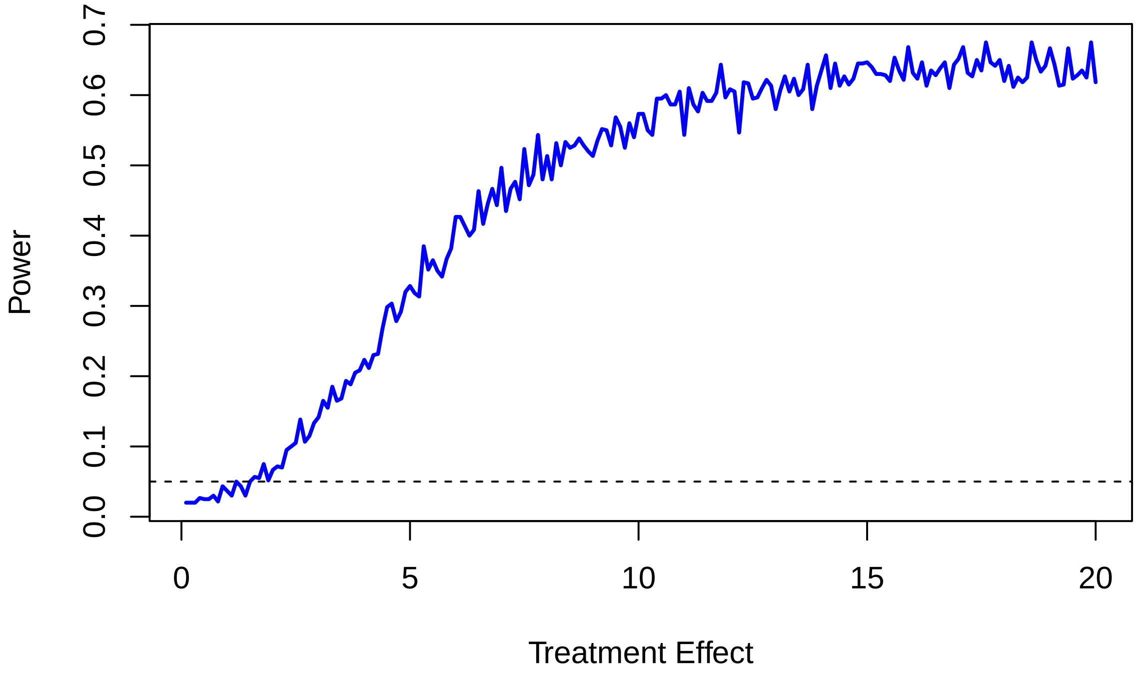
```
# Print the treatment effect after which the power is greater than 0.05
treatment_effects[which(power > 0.05)[1:10]]
```

```
##  [1] 1.6 1.7 1.8 1.9 2.0 2.1 2.2 2.3 2.4 2.5
```

```
# Print the treatment effect after which the power is greater than 0.6
treatment_effects[which(power > 0.6)[1:10]]
```

```
##  [1] 10.9 11.1 11.4 11.7 11.8 12.0 12.1 12.3 12.4 12.7
```

```
# Plot the power
par(mfrow = c(1, 1))
plot(treatment_effects, power, type = "l", xlab = "Treatment Effect", ylab = "Power", col = "blue", lwd
abline(h = 0.05, lty = 2) # Add a horizontal line at 0.05
```

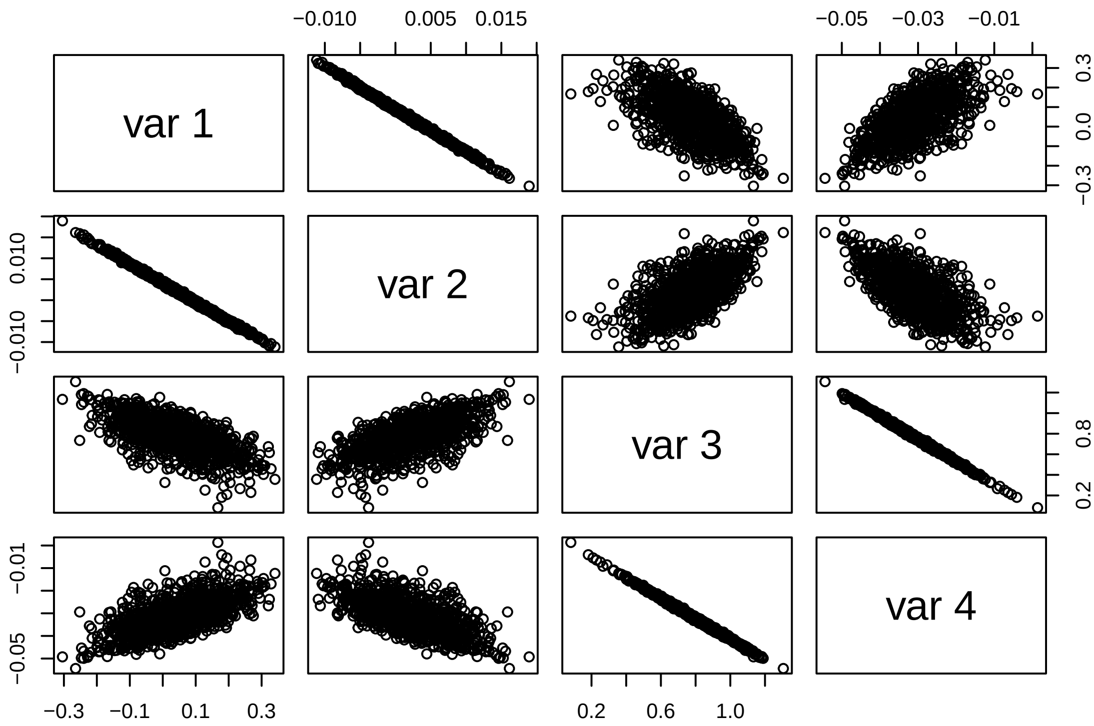


9 Simulation to assess the suitability of the commonly used asymptotic approximations for the sampling distribution of the maximum likelihood esimators and scaled deviance in this problem

```
# Set the seed for reproducibility
set.seed(604859)

# Pick the model
model <- fit

# Set the number of simulations
num_sim <- 1000

# Extract some useful quantities
num_subjects <- nrow(dat)
num_coeffs = model$rank
dof = num_subjects - num_coeffs
coeff_names <- rownames(summary(model)$coefficients)

# Create empty matrices to store the results
coeffs_sim <- matrix(nrow = num_sim, ncol=num_coeffs)
dev_sim <- numeric(num_sim)
```

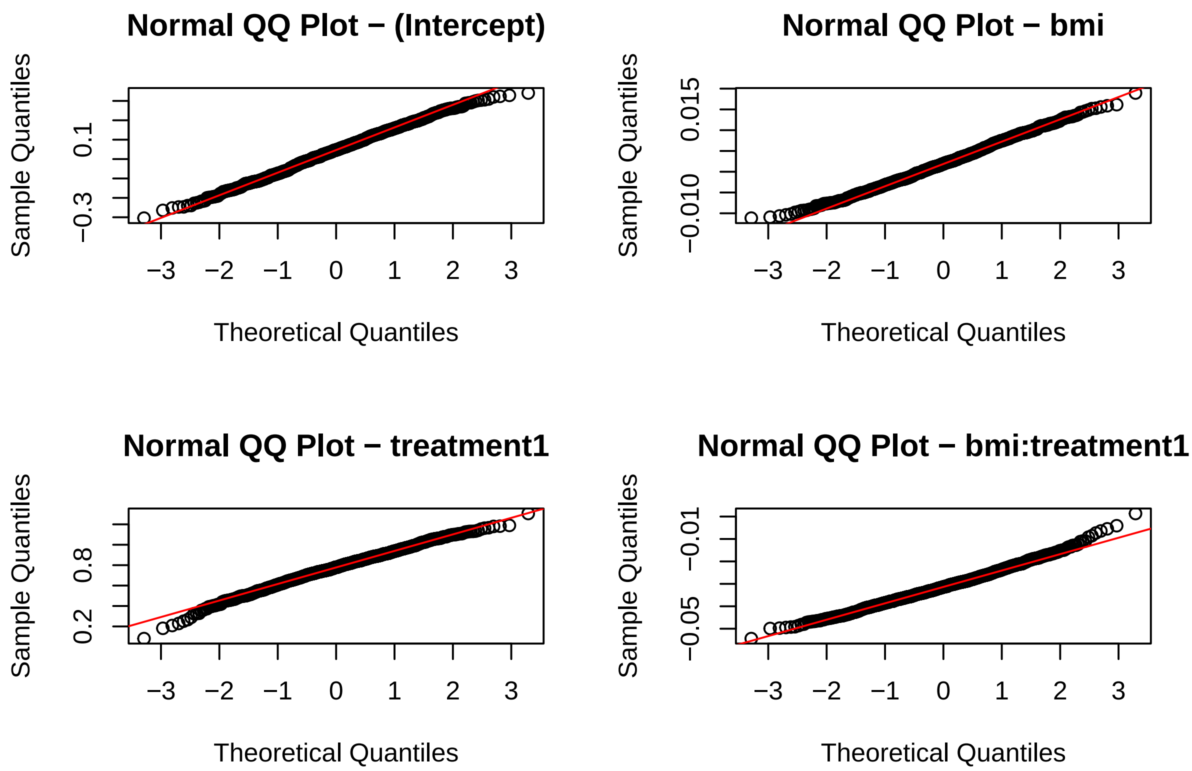
```r
# Run the simulations
for (i in 1:num_sim) {
  # Simulate the output variable
  diff_sim <- unlist(simulate(model))

  # Fit the model to the simulated data
  form = diff_sim ~ bmi * treatment
  fit_sim <- glm(form, family = Gamma(link = "inverse"), data = dat)
  summary_fit_sim <- summary(fit_sim)

  # Get the dispersion parameter
  dispersion <- summary_fit_sim$dispersion

  # Store the coefficients
  coeffs_sim[i,] <- coef(fit_sim)
  # Store the scaled deviance
  dev_sim[i] <- summary_fit_sim$deviance / dispersion
}
```

```r
# Plot the pairs of coefficients
pairs(coeffs_sim)
```




```r
# Compare the residuals to the normal distribution
par(mfrow = c(2,ifelse(num_coeffs %% 2 == 0, num_coeffs / 2, (num_coeffs + 1) / 2)))
for(j in 1:num_coeffs){
  qqnorm(coeffs_sim[,j], main=paste("Normal QQ Plot -", coeff_names[j]))
```

```
  qqline(coeffs_sim[,j], col = "red")
}
```




```
# Perform the Kolmogorov-Smirnov test for the residuals
for(j in 1:num_coeffs){
  standardized_residuals <- (coeffs_sim[,j] - mean(coeffs_sim[,j])) / sd(coeffs_sim[,j])
  print(paste("Kolmogorov-Smirnov test p-value for coefficients",
              coeff_names[j], ": ",
              ks.test(standardized_residuals, "pnorm")$p.value))
}
```

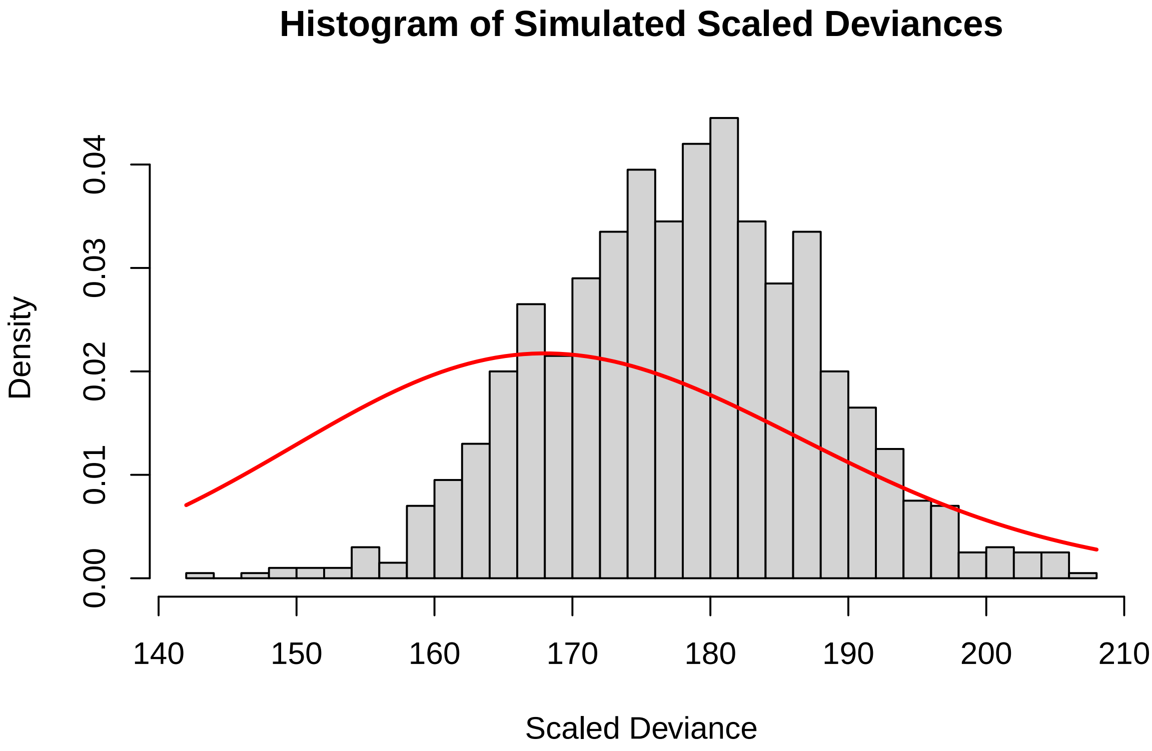
```
## [1] "Kolmogorov-Smirnov test p-value for coefficients (Intercept) :  0.741003637019485"
## [1] "Kolmogorov-Smirnov test p-value for coefficients bmi :  0.69453565351338"
## [1] "Kolmogorov-Smirnov test p-value for coefficients treatment1 :  0.319987566817416"
## [1] "Kolmogorov-Smirnov test p-value for coefficients bmi:treatment1 :  0.277589941502756"
```

```
# Perform a test from the MVN package in order to check the multivariate normality
mvn(coeffs_sim, mvnTest = "mardia", desc=TRUE)$multivariateNormality
```

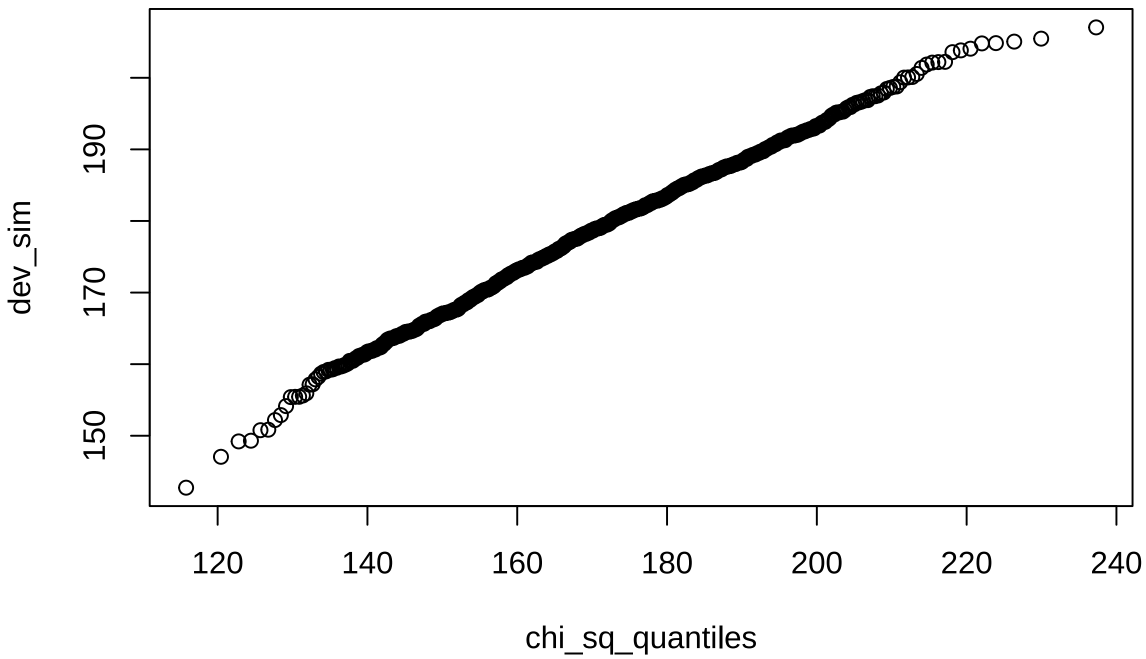
```
##              Test        Statistic              p value Result
## 1 Mardia Skewness 191.060217749739 6.54281112386184e-30     NO
## 2 Mardia Kurtosis 3.86643419308828 0.000110438245979338     NO
## 3             MVN             <NA>                 <NA>     NO
```

```
# Plot the histogram of simulated deviances
par(mfrow = c(1, 1))
hist(dev_sim, breaks=40, probability = TRUE,
     main="Histogram of Simulated Scaled Deviances",
```

```
     xlab="Scaled Deviance")
# Overlay the chi-squared distribution
curve(dchisq(x, df=dof), add=TRUE, col="red", lwd=2)
```




```
# Compare the scaled deviance to the chi-squared distribution
par(mfrow = c(1, 1))
chi_sq_quantiles <- qchisq(ppoints(dev_sim), df = dof)
qqplot(chi_sq_quantiles, dev_sim)
qqline(dev_sim, col = "red")
```




```
# Perform a Kolmogorov-Smirnov test
ks.test(dev_sim, "pchisq", dof)
```

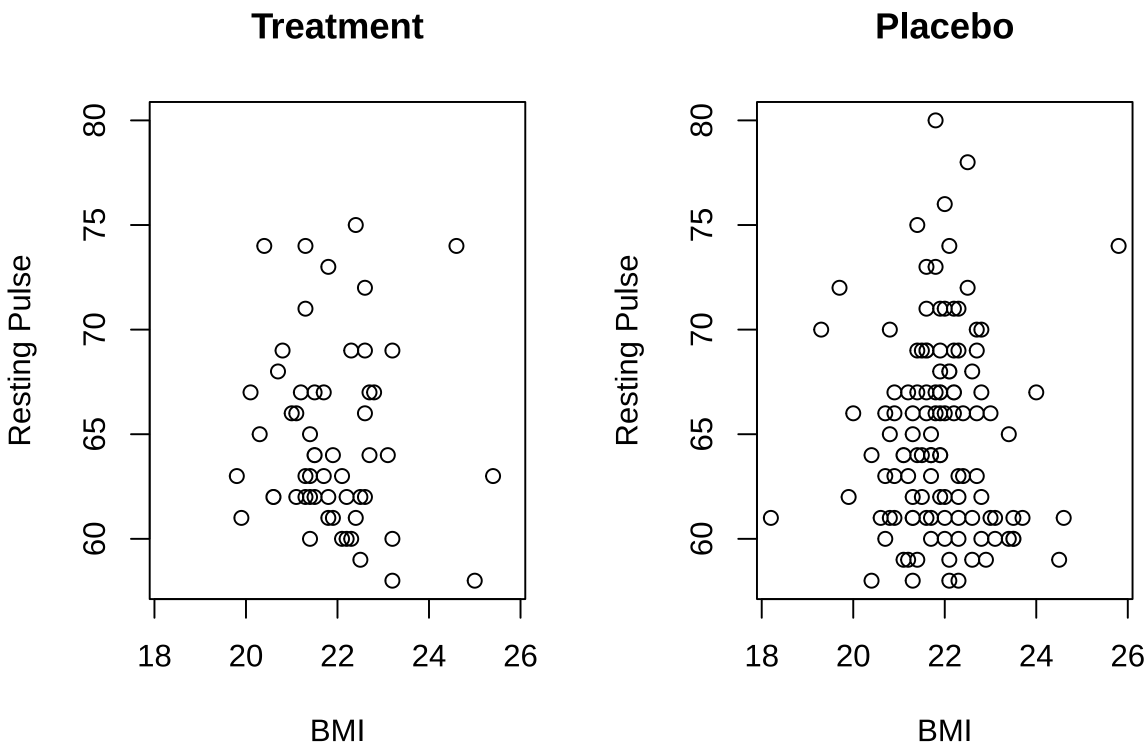
```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  dev_sim
## D = 0.31687, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

10 Relationship between BMI and resting pulse

```
# Print the correlation between BMI and resting pulse
cor(bmi, rest_pulse)
```

```
## [1] -0.02819013
```

```
# Print the resting pulse as a function of BMI
par(mfrow = c(1, 2))
plot(bmi[treatment == 0], rest_pulse[treatment == 0],
     xlab = "BMI", ylab = "Resting Pulse",
     main = "Treatment", xlim = range(bmi), ylim = range(rest_pulse))
plot(bmi[treatment == 1], rest_pulse[treatment == 1],
     xlab = "BMI", ylab = "Resting Pulse",
     main = "Placebo", xlim = range(bmi), ylim = range(rest_pulse))
```


Treatment
Placebo
Resting Pulse
BMI
60
65
70
75
80
18
20
22
24
26